Lab 07

Configuring Network Firewalls Using ZPF and ACL

Name: Syed Saad Imran	Student ID: 09370

7.1 Objective

The Objectives of this lab are:

- Configure a zone-based policy (ZPF) firewall on R3.
- Configure a basic firewall
- Comfigure a router based ACL

ZPFs are the latest development in the evolution of Cisco firewall technologies. In this activity, you will configure a basic ZPF on an edge router R3 that allows internal hosts access to external resources and blocks external hosts from accessing internal resources. You will then verify firewall functionality from internal and external hosts.

The routers have been pre-configured with the following:

- Console password: **ciscoconpa55**
- Password for vty lines: **ciscovtypa55**
- Enable password: **ciscoenpa55**
- Host names and IP addressing
- Local username and password: **Admin** / **Adminpa55**
- Static routing

Topology

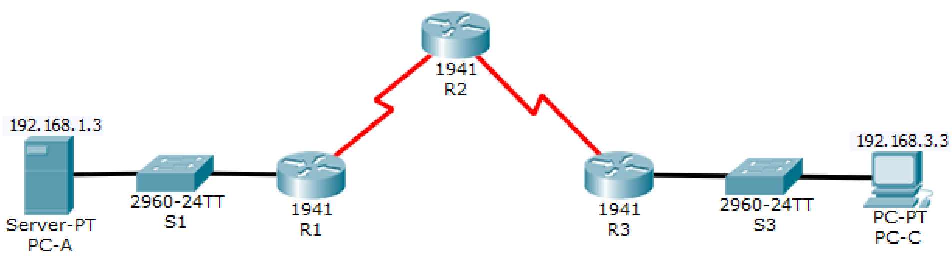


Figure 1: Topology

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Table 1: Addressing Table

Task 1: Configuring a Zone-Based Policy Firewall (ZPF)

<u>Verify Basic Network Connectivity</u>

Verify network connectivity prior to configuring the zone-based-policy firewall.

1. From the PC-A command prompt, ping PC-C at 192.168.3.3

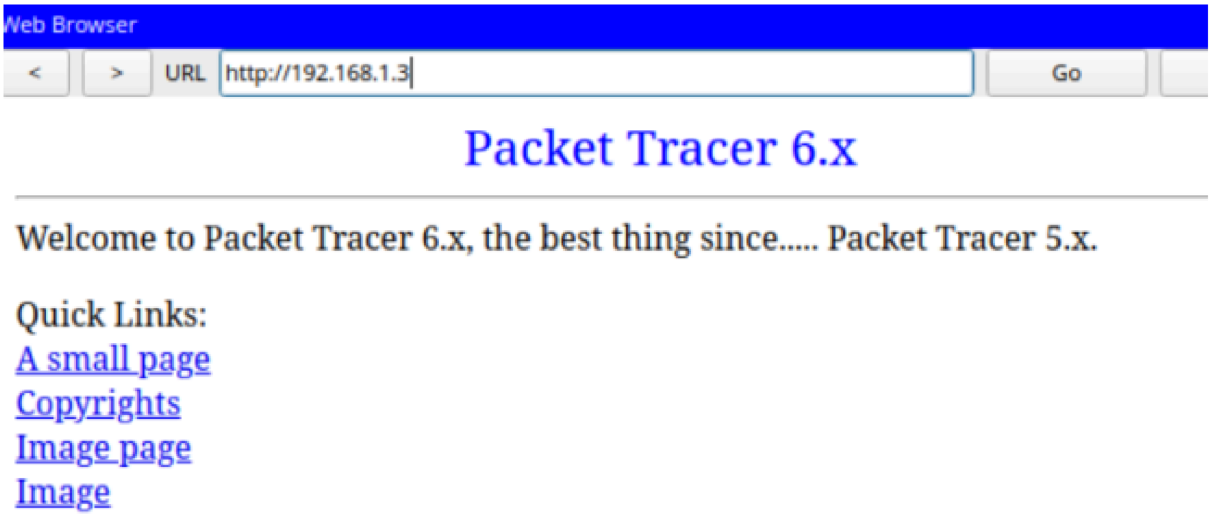
```
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time=12ms TTL=125
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 12ms, Average = 4ms

C:\>
```

2. Access R2 using SSH.
   a. From the **PC-C** command prompt, SSH to the S0/0/1 interface on **R2** at **10.2.2.2**. Use the username **Admin** and password **Adminpa55** to log in.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l Admin 10.2.2.2

Password:


R2#
```

   b. Exit the SSH session.

3. From PC-C, open a web browser to the PC-A server.
   a. Click the **Desktop** tab and then click the **Web Browser** application. Enter the **PC-A** IP address **192.168.1.3** as the URL. The Packet Tracer welcome page from the web server should be displayed.



   b. Close the browser on **PC-C.**

Create the Firewall Zones on R3

Note: For all configuration tasks, be sure to use the exact names as specified.

1. Enable the Security Technology package.
   a. On R3, issue the show version command to view the Technology Package license information.

```
License Info:

License UDI:

-------------------------------------------------
Device#   PID                   SN
-------------------------------------------------
*0        CISCO1941/K9          FTX152446W3


Technology Package License Information for Module:'c1900'

----------------------------------------------------------------
Technology    Technology-package           Technology-package
              Current       Type           Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent      ipbasek9
security      disable       None           securityk9
data          disable       None           None

Configuration register is 0x2102
```

b. If the Security Technology package has not been enabled, use the following command to enable the package.

R3(config)# **license boot module c1900 technology-package securityk9**

c. Accept the end-user license agreement.
d. Save the running-config and reload the router to enable the security license.
e. Verify that the Security Technology package has been enabled by using the show version command.

```
License Info:

License UDI:

-------------------------------------------------
Device#   PID                   SN
-------------------------------------------------
*0        CISCO1941/K9          FTX152446W3


Technology Package License Information for Module:'c1900'

----------------------------------------------------------------
Technology    Technology-package           Technology-package
              Current       Type           Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent      ipbasek9
security      securityk9    Evaluation     securityk9
data          disable       None           None

Configuration register is 0x2102
```

2. Create an internal zone.

   Use the zone security command to create a zone named **IN-ZONE.**

   R3(config)# **zone security IN-ZONE**

   R3(config-sec-zone) **exit**

3. Create an external zone.

   Use the zone security command to create a zone named **OUT-ZONE.**

   R3(config-sec-zone)# **zone security OUT-ZONE**

   R3(config-sec-zone)# **exit**

```
R3(config)#zone security IN-ZONE
R3(config-sec-zone)#exit
R3(config)#zone security OUT-ZONE
R3(config-sec-zone)#exit
R3(config)#
```

Identify Traffic Using a Class-Map

1. Create an ACL that defines internal traffic.

   Use the **access-list** command to create extended ACL **101** to permit all IP protocols from the **192.168.3.0/24** source network to any destination.

   R3(config)# **access-list 101 permit ip 192.168.3.0 0.0.0.255 any**

```
R3(config)#access-list 101 permit ip 192.168.3.0 0.0.0.255 any
R3(config)#
```

2. Create a class map referencing the internal traffic ACL.

   Use the class-map type inspect command with the match-all option to create a class map named **IN-NET-CLASS-MAP.** Use the **match access-group** command to match ACL **101**.

   R3(config)# **class-map type inspect match-all IN-NET-CLASS-MAP**

   R3(config-cmap)# **match access-group 101**

   R3(config-cmap)# **exit**

```
R3(config)#class-map type inspect match-all IN-NET-CLASS-MAP
R3(config-cmap)#match access-group 101
R3(config-cmap)#exit
R3(config)#
```

Specify Firewall Policies

1. Create a policy map to determine what to do with matched traffic.

   Use the **policy-map type inspect** command and create a policy map named **IN-2-OUT-PMAP**.

   ```
   R3(config)# policy-map type inspect IN-2-OUT-PMAP
   ```

2. Specify a class type of inspect and reference class map IN-NET-CLASS-MAP.

   ```
   R3(config-pmap)# class type inspect IN-NET-CLASS-MAP
   ```

3. Specify the action of inspect for this policy map.

   The use of the **inspect** command invokes context-based access control (other options include pass and drop).

   ```
   R3(config-pmap-c)# inspect
   ```

   %No specific protocol configured in class IN-NET-CLASS-MAP for inspection. All protocols will be inspected.

   Issue the **exit** command twice to leave **config-pmap-c** mode and return to **config** mode.

   ```
   R3(config-pmap-c)# exit
   ```

   ```
   R3(config-pmap)# exit
   ```

```
R3(config)#policy-map type inspect IN-2-OUT-PMAP
R3(config-pmap)#class type inspect IN-NET-CLASS-MAP
R3(config-pmap-c)#inspect
%No specific protocol configured in class IN-NET-CLASS-MAP for inspection. All protocols will be
inspected
R3(config-pmap-c)#exit
R3(config-pmap)#exit
R3(config)#
```

Apply Firewall Policies

1. **Create a pair of zones.**

   Using the **zone-pair security** command, create a zone pair named **IN-2-OUT-ZPAIR**. Specify the source and destination zones that were created in Task 1.

R3(config)# **zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-ZONE**

2. **Specify the policy map for handling the traffic between the two zones.**

   Attach a policy-map and its associated actions to the zone pair using the **service-policy type inspect** command and reference the policy map previously created, **IN-2-OUT-PMAP**.

   R3(config-sec-zone-pair)# **service-policy type inspect IN-2-OUT-PMAP**

   R3(config-sec-zone-pair)# **exit**

   R3(config)#

```
R3(config)#zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-ZONE
R3(config-sec-zone-pair)#service-policy type inspect IN-2-OUT-PMAP
R3(config-sec-zone-pair)#exit
```

3. **Assign interfaces to the appropriate security zones.**

   Use the zone-member security command in interface configuration mode to assign G0/1 to IN-ZONE and S0/0/1 to OUT-ZONE.

   R3(config)# **interface g0/1**

   R3(config-if)# **zone-member security IN-ZONE**

   R3(config-if)# **exit**

   R3(config)# **interface s0/0/1**

   R3(config-if)# **zone-member security OUT-ZONE**

   R3(config-if)# **exit**

```
R3(config-if)#zone-member security IN-ZONE
R3(config-if)#exit
R3(config)#interface s0/0/1
R3(config-if)#zone-member security OUT-ZONE
R3(config-if)#exit
R3(config)#
```

4. Copy the running configuration to the startup configuration.

<u>Test Firewall Functionality from IN-ZONE to OUT-ZONE</u>

Verify that internal hosts can still access external resources after configuring the ZPF.

1. From internal PC-C, ping the external PC-A server.

From the PC-C command prompt, ping PC-A at 192.168.1.3. The ping should succeed.

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=3ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=13ms TTL=125

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 13ms, Average = 5ms

C:\>
```

2. From internal PC-C, SSH to the R2 S0/0/1 interface.

a. From the **PC-C** command prompt, SSH to **R2** at 10.2.2.2. Use the username **Admin** and the password **Adminpa55** to access R2. The SSH session should succeed.

```
C:\>ssh -l Admin 10.2.2.2

Password:
% Login invalid


Password:



R2#
```

b. While the SSH session is active, issue the command **show policy-map type inspect zone-pair sessions** on **R3** to view established sessions.

```
R3#show policy-map type inspect zone-pair sessions

policy exists on zp IN-2-OUT-ZPAIR
 Zone-pair: IN-2-OUT-ZPAIR

  Service-policy inspect : IN-2-OUT-PMAP

    Class-map: IN-NET-CLASS-MAP (match-all)
      Match: access-group 101
      Inspect

        Number of Established Sessions = 1
        Established Sessions
         Session 2492978192 (192.168.3.3:1026)=>(10.2.2.2:22) tcp SIS_OPEN/TCP_ESTAB
          Created 00:01:50, Last heard  00:00:51
          Bytes sent (initiator:responder) [5762:3541]
    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
R3#
```

What is the source IP address and port number?

Source ip 192.168.3.3 and port number 1026

What is the destination IP address and port number?

Destination ip 10.2.2.2 and port number 22

3. From PC-C, exit the SSH session on R2 and close the command prompt window.
4. From internal PC-C, open a web browser to the PC-A server web page.

   Enter the server IP address **192.168.1.3** in the browser URL field, and click **Go**. The HTTP session should succeed. While the HTTP session is active, issue the command **show policy-map type inspect zone-pair sessions** on **R3** to view established sessions.

   Note: If the HTTP session times out before you execute the command on **R3**, you will have to click the **Go** button on **PC-C** to generate a session between **PC-C** and **PC-A**.

```
R3#show policy-map type inspect zone-pair sessions

policy exists on zp IN-2-OUT-ZPAIR
 Zone-pair: IN-2-OUT-ZPAIR

  Service-policy inspect : IN-2-OUT-PMAP

    Class-map: IN-NET-CLASS-MAP (match-all)
      Match: access-group 101
      Inspect

        Number of Established Sessions = 1
        Established Sessions
         Session 415500784 (192.168.3.3:1035)=>(192.168.1.3:80) tcp SIS_OPEN/TCP_ESTAB
          Created 00:00:04, Last heard  00:00:04
          Bytes sent (initiator:responder) [284:552]
    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
```

   What is the source IP address and port number?

   Source ip 192.168.3.3 and port number 1035

   What is the destination IP address and port number?

   Destination ip 192.168.1.3 and port number 80

5. Close the browser on PC-C.

Test Firewall Functionality from OUT-ZONE to IN-ZONE

Verify that external hosts CANNOT access internal resources after configuring the ZPF.

1. From the PC-A server command prompt, ping PC-C.

   From the **PC-A** command prompt, ping **PC-C** at 192.168.3.3. The ping should fail.

```
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

2. From R2, ping PC-C.

   From R2, ping PC-C at 192.168.3.3. The ping should fail.

```
R2#ping 192.168.3.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.3, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

3. Check results.

   Your completion percentage should be 100%. Click **Check Results** to see feedback and verification of which required components have been completed.

Task 2: Basic Firewall Configuration

Topology

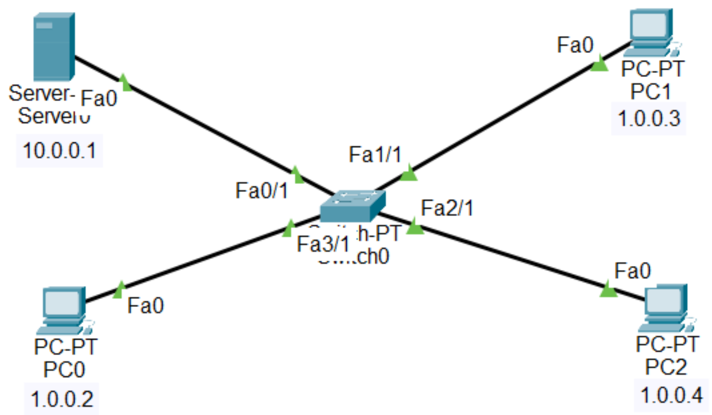


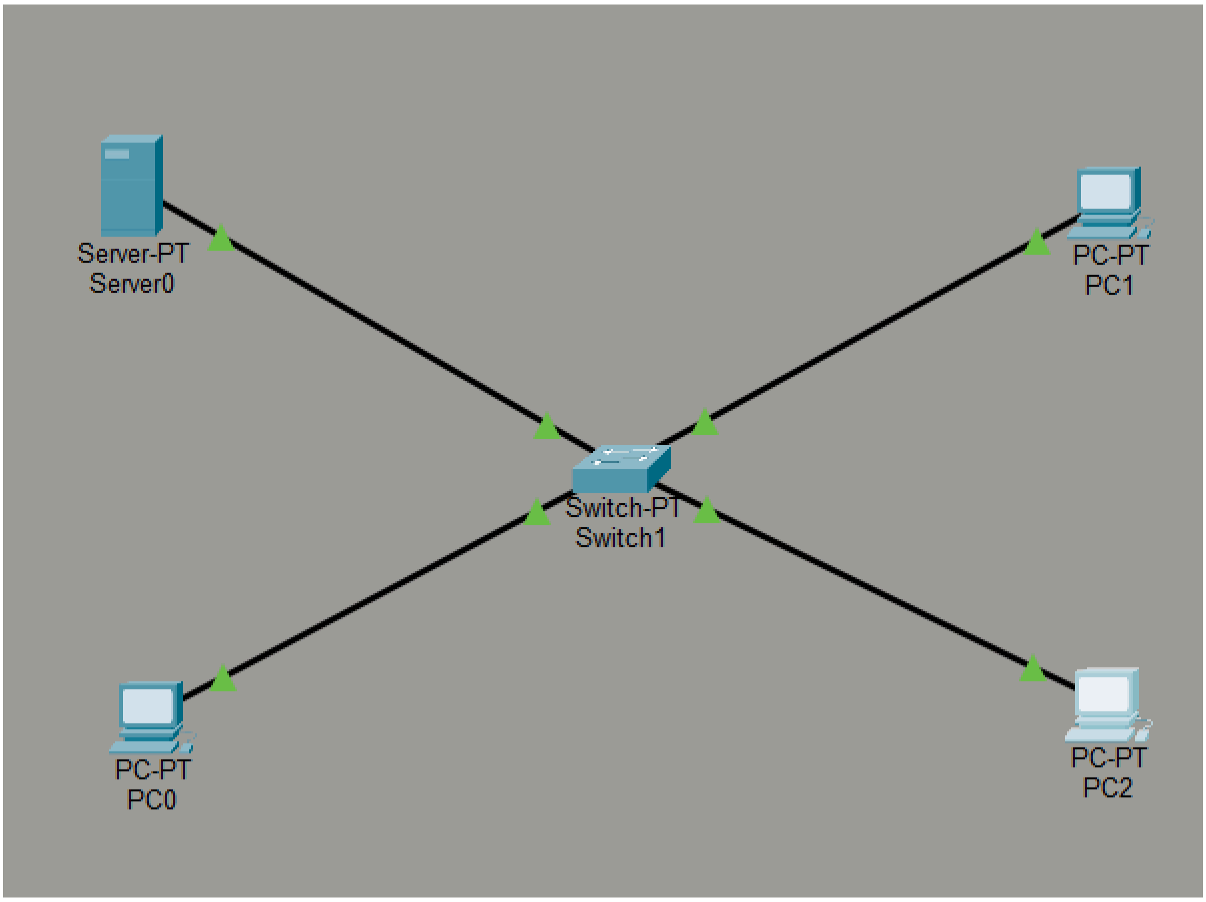
Figure 2: Topology for Task 2

Addressing Table

Device	IP Address	Subnet Mask
Server	1.0.0.1	255.0.0.0
PC0	1.0.0.2	255.0.0.0
PC1	1.0.0.3	255.0.0.0
PC2	1.0.0.4	255.0.0.0

Table 2: Addressing Table for Task 2

Construct the topology as shown in the figure above. Assign IP addresses and subnet masks. Attach screenshot of the topology.



1) Configure the firewall server. Attach Screenshot.
   a) Click on server0, then go to the desktop.
   b) Click on Firewall IPv4, and **turn on the service.**
   c) First deny the icmp protocol and set remote IP to 0.0.0.0 and remote wildcard mask to 255.255.255.255.
   d) Allow the IP protocol and set remote IP to 0.0.0.0 and remote wildcard mask to 255.255.255.255.

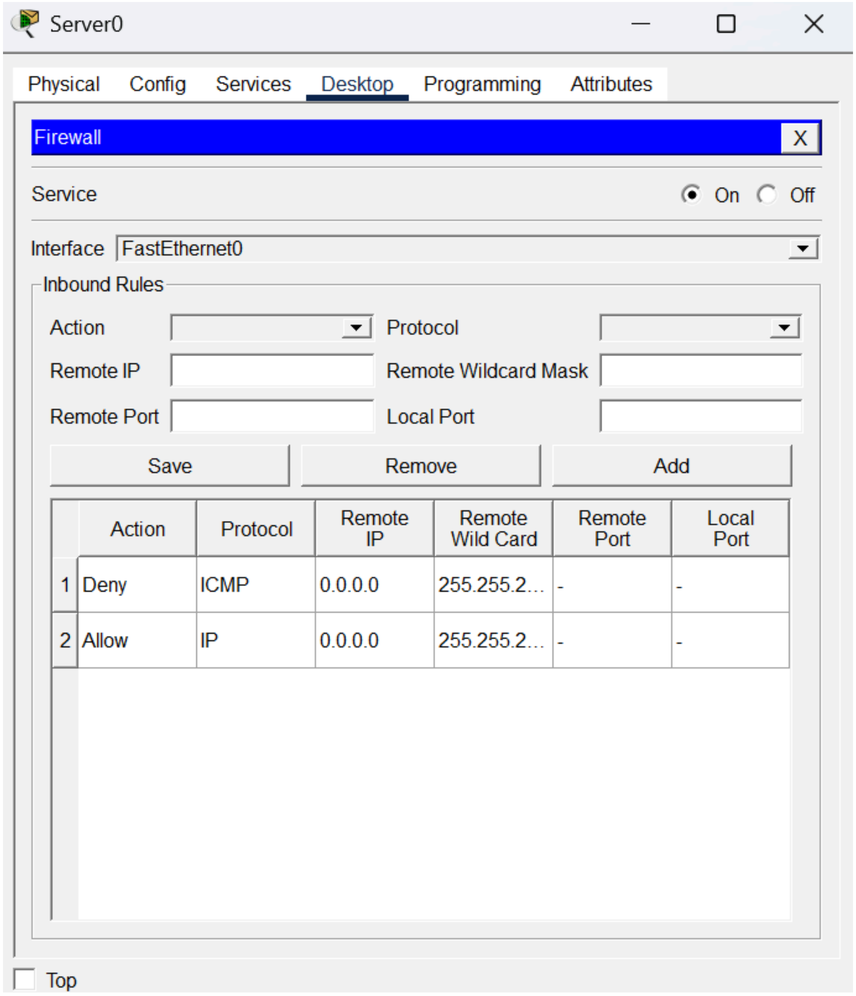
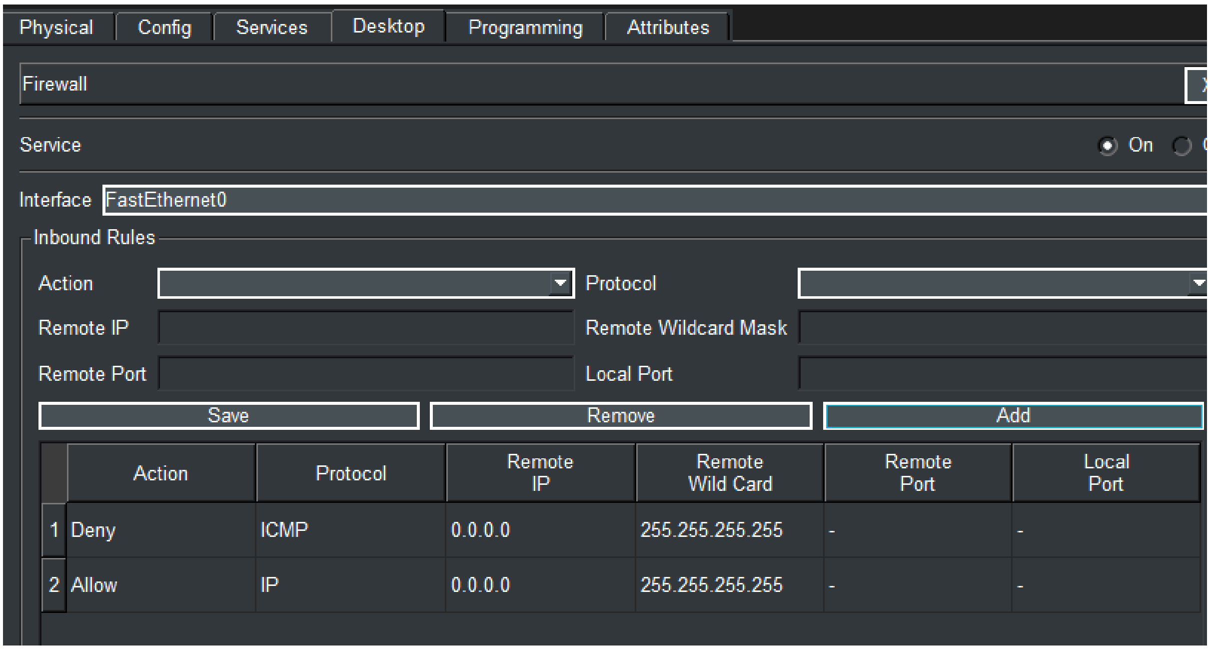
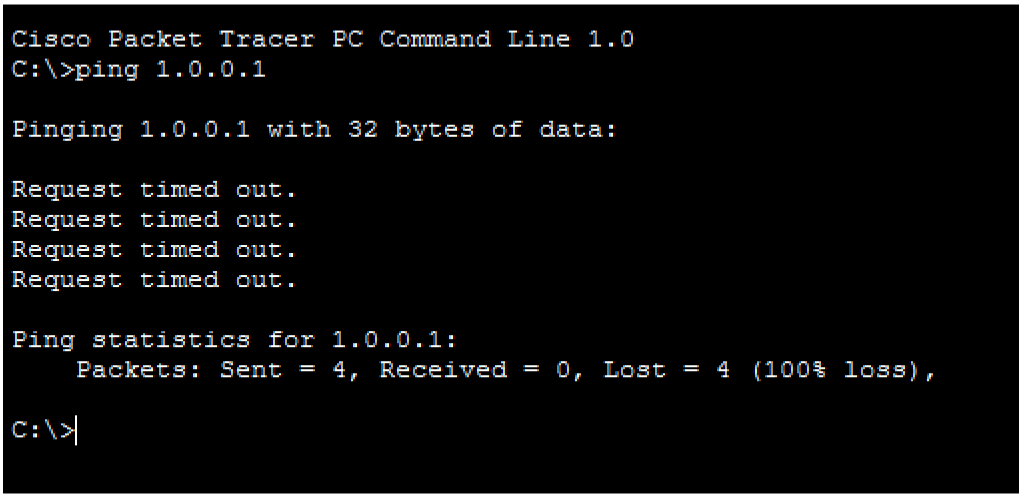


Figure 3: Firewall Configuration

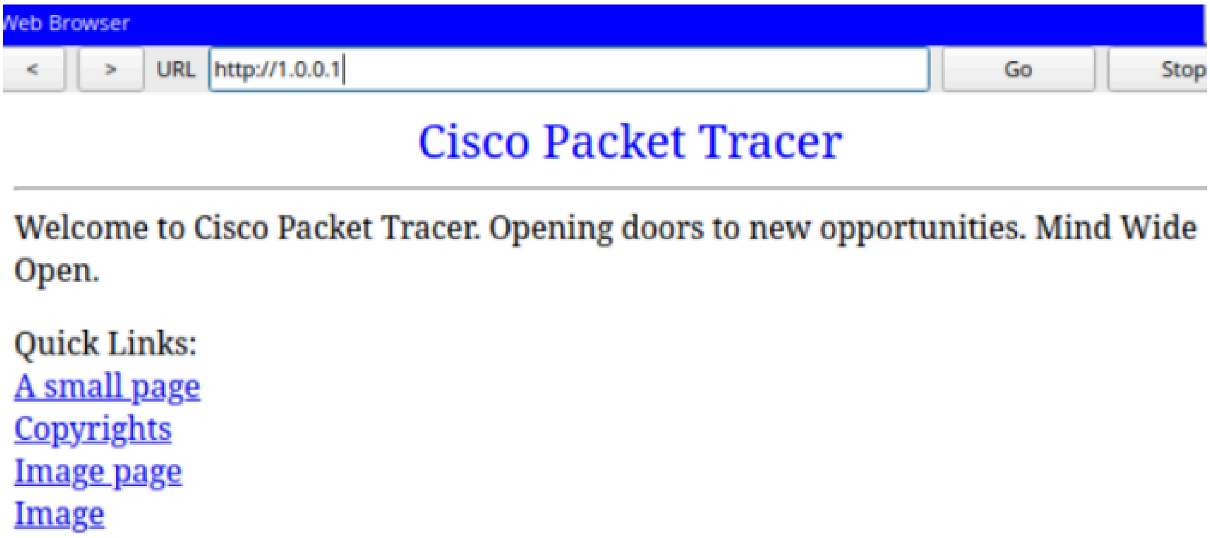


2) Verify the network by pinging the IP address of any PC. Attach Screenshot.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 1.0.0.1

Pinging 1.0.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 1.0.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

3) Check the web browser by entering the IP address in the URL. Attach Screenshot.


Web Browser
<
>
URL
http://1.0.0.1
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image

Task 3: Implementing Firewall Using Router

Addressing Table:

Device	IP Address	Subnet Mask	Default Gateway
Server	2.0.0.2	255.0.0.0	2.0.0.1
PC0	1.0.0.2	255.0.0.0	1.0.0.5
PC1	1.0.0.3	255.0.0.0	1.0.0.5
PC2	1.0.0.4	255.0.0.0	1.0.0.5
Router G0/1	1.0.0.5	255.0.0.0	N/A
Router G0/0	2.0.0.1	255.0.0.0	N/A

Table 3: Topology for Task 3

1) Modify the topology from Task 1 by inserting a **router** between the PCs and the server. Attach Screenshot.
   a) Use a **switch** to connect the PCs to the router.
   b) Connect the router's **GigabitEthernet0/0** interface to the server.
2) Assign IP addresses and default gateways according to the addressing table provided above.
3) Instead of using the server's firewall, implement the security rules directly on the router:
   Router(config)# access-list 101 deny icmp any host 2.0.0.2
   Router(config)#access-list 101 permit ip any any

```
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#access-list 101 deny icmp any host 2.0.0.2
Router(config)#access-list 101 permit ip any any
Router(config)#interface g0/1
Router(config-if)#ip access-group 101 in
Router(config-if)#exit
```

4) Apply ACL to PC facing interface.
5) Verify network connectivity by pinging the server. The ping requests should fail because the router is configured to deny ICMP traffic. Attach Screenshot.

```
C:\>ping 2.0.0.2

Pinging 2.0.0.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 2.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

6) Test web access by entering the server's IP address in the web browser. Attach screenshot.


Web Browser
<
>
URL
http://2.0.0.2
Go
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Min
Open.
Quick Links:
A small page
Copyrights
Image page
Image

Assessment Rubric
Lab 07
Configuring Network Firewalls Using ZPF and ACL

Name:	**Student ID:**

Points Distribution

Task No.	LR 2 Simulation	LR5 Results/Plots	LR9 Report
Task 1	/35	/15	
Task 2	/10	/5	
Task 3	/10	/5	
Total	/55	/25	/10
CLO Mapped	CLO 3	CLO 3	CLO3

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission	/10	CLO 3

CLO	Total Points	Points Obtained
3	100	
Total	**100**	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR2	**Program/Code / Simulation Model/ Network Model**	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR9	**Report**	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.